

实验报告

```
terminal
[ 25%] Building C object libs/ggml/src/CMakeFiles/ggml.dir/ggml.c.o
[ 31%] Building C object libs/ggml/src/CMakeFiles/ggml.dir/ggml-alloc.c.o
[ 37%] Building C object libs/ggml/src/CMakeFiles/ggml.dir/ggml-backend.c.o
[ 43%] Building C object libs/ggml/src/CMakeFiles/ggml.dir/ggml-quants.c.o
[ 50%] Building C object libs/ggml/src/CMakeFiles/ggml.dir/ggml-aarch64.c.o
[ 56%] Linking C shared library libggml.so
gmake[2]: warning: Clock skew detected. Your build may be incomplete.
[ 56%] Built target ggml
[ 62%] Building CXX object libs/llama_tokenizer/CMakeFiles/llama_tokenizer.dir/llama_vocab.cpp.o
[ 68%] Building CXX object libs/llama_tokenizer/CMakeFiles/llama_tokenizer.dir/unicode_data.cpp.o
[ 75%] Building CXX object libs/llama_tokenizer/CMakeFiles/llama_tokenizer.dir/unicode.cpp.o
[ 81%] Linking CXX static library libllama_tokenizer.a
[ 81%] Built target llama_tokenizer
gmake[2]: Warning: File 'src/CMakeFiles/run.dir/compiler_depend.make' has modification time 0.53 s in the future
[ 87%] Building CXX object src/CMakeFiles/run.dir/main.cpp.o
[ 93%] Building CXX object src/CMakeFiles/run.dir/core.cpp.o
/mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/core.cpp: In member function 'float* sep::Transformer::forward(
int, int)':
/mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/core.cpp:168:16: warning: address of local variable 'logits' r
eturned [-Wreturn-local-addr]
   168 |         return logits;
       |         ^
/mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/core.cpp:149:15: note: declared here
   149 |         float logits[p->vocab_size];
       |         ^
[100%] Linking CXX executable run
gmake[2]: warning: Clock skew detected. Your build may be incomplete.
[100%] Built target run
administrator@PC-20250120DLBD:/mnt/c/Users/Administrator/Desktop/sep/debug2-handout$ |
```

1.

使用 Asan 进行 debug，在输入 `cmake --build build --config RelWithDebInfo` 后报错信息指出了 **"address of local variable 'logits' returned"**，这个警告说明在函数 `Transformer::forward()` 中返回了局部变量 `logits` 的地址，访问这些地址会导致未定义行为。在报错中提到的是

`float logits[p->vocab_size];` 这一行代码，而后在函数结束时 `return logits;`

这会导致返回一个指向栈内存的指针，这个指针在 `forward()` 函数结束后失效。

解决方案：改为 `float *logits = s->logits;`

返回一个类成员变量的指针，而不是局部变量。这样，`RunState` 对象销毁时，`logits` 的内存会被正确地释放，避免了内存泄漏。

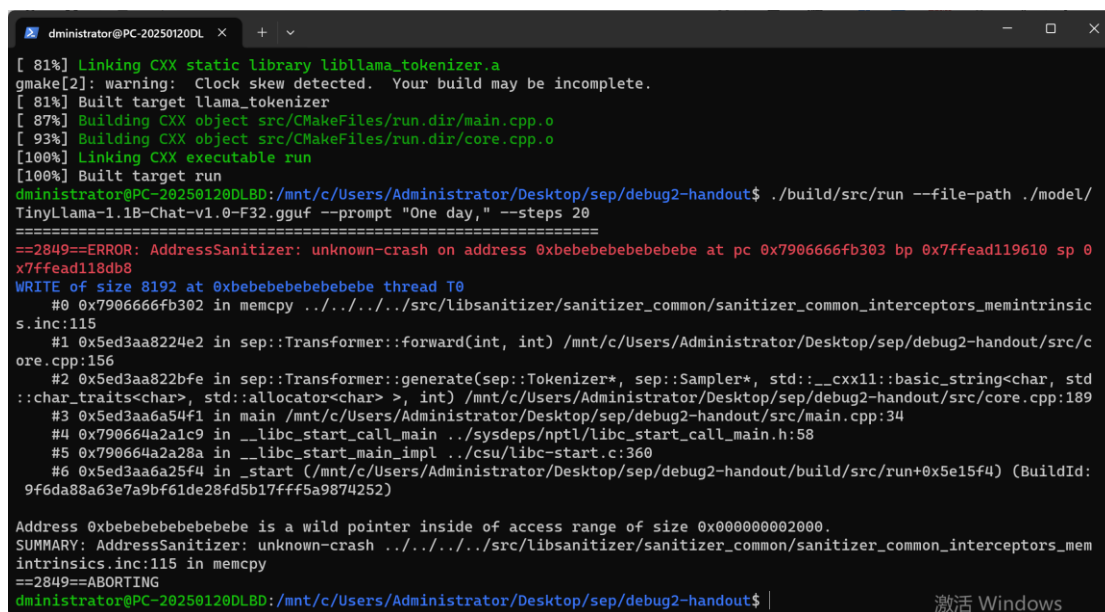
2. 通过观察代码逻辑发现，两个析构函数里内存没有正确释放，因

此在析构函数中加入了正确的内存释放操作，以避免内存泄漏。

```
3. RunState::~~RunState() { delete[] x; delete[] xb; delete[] xb2;
    delete[] hb; delete[] hb2; delete[] q; delete[] key_cache; delete[]
    value_cache; delete[] att; delete[] logits; }
4. Transformer::~~Transformer() { delete state; delete weight; delete
    config; gguf_free(gguf_ctx_); }
```

3. 通过 gdb 调试，在 RunState::RunState 处设置断点并检查各个变量的值后发现，config->dim 的值为 4294956304，而 config->n_layers 的值为 1432071583，已经超出 65535 (uint_t16 的范围)，所以将各变量类型改为 size_t 以确保不发生整数溢出。

4.



```
dministrator@PC-20250120DL x + v
[ 81%] Linking CXX static library libllama_tokenizer.a
gmake[2]: warning: Clock skew detected. Your build may be incomplete.
[ 81%] Built target llama_tokenizer
[ 87%] Building CXX object src/CMakeFiles/run.dir/main.cpp.o
[ 93%] Building CXX object src/CMakeFiles/run.dir/core.cpp.o
[100%] Linking CXX executable run
[100%] Built target run
dministrator@PC-20250120DLBD:/mnt/c/Users/Administrator/Desktop/sep/debug2-handout$ ./build/src/run --file-path ./model/
TinyLlama-1.1B-Chat-v1.0-F32.gguf --prompt "One day," --steps 20
=====
==2849==ERROR: AddressSanitizer: unknown-crash on address 0xbebebebebebe at pc 0x7906666fb303 bp 0x7ffead119610 sp 0
x7ffead118db8
WRITE of size 8192 at 0xbebebebebebebebebebe thread T0
#0 0x7906666fb302 in memcpy ../../../../src/libsanitizer/sanitizer_common/sanitizer_common_interceptors_memintrinsic
s.inc:115
#1 0x5ed3aa8224e2 in sep::Transformer::forward(int, int) /mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/c
ore.cpp:156
#2 0x5ed3aa822bfe in sep::Transformer::generate(sep::Tokenizer*, sep::Sampler*, std::__cxx11::basic_string<char, std
::char_traits<char>, std::allocator<char> >, int) /mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/core.cpp:189
#3 0x5ed3aa6a54f1 in main /mnt/c/Users/Administrator/Desktop/sep/debug2-handout/src/main.cpp:34
#4 0x790664a2a1c9 in __libc_start_call_main ../sysdeps/nptl/libc_start_call_main.h:58
#5 0x790664a2a28a in __libc_start_main_impl ../csu/libc-start.c:360
#6 0x5ed3aa6a25f4 in _start (/mnt/c/Users/Administrator/Desktop/sep/debug2-handout/build/src/run+0x5e15f4) (BuildId:
9f6da88a63e7a9bf61de28fd5b17fff5a9874252)

Address 0xbebebebebebebebebebe is a wild pointer inside of access range of size 0x000000002000.
SUMMARY: AddressSanitizer: unknown-crash ../../../../src/libsanitizer/sanitizer_common/sanitizer_common_interceptors_mem
intrinsics.inc:115 in memcpy
==2849==ABORTING
dministrator@PC-20250120DLBD:/mnt/c/Users/Administrator/Desktop/sep/debug2-handout$ | 激活 Windows
```

通过 Asan 调试发现 forward 函数中 memcpy(s->x, ...)这一行访问了已被释放 / 未正确分配的内存，错误发生在 sep::Transformer::forward 的 core.cpp:156 行 → 对应代码是 memcpy(s->x, content_row, dim * sizeof(float)); 说明问题出在 s->x 这个指针：它指向的内存已经被释放，或者根本没正确分配。结合你之前的代码，这个问题的直接原因是 RunState 中的 x 数组未正确初始化。看之前的代码，早期 RunState 构造函数是没有分配 x 的，所以加上分配 x 的代码 x = new float[dim]; 同时检查 attention 函数，同时发现 RunState 对象被提前释放，后续 forward 函数访问 s->x，就是访问已释放的内存，所以，删去 delete s;这一行。此后程序可以正常运行